Question 1 on Flexible AC Transmission System (FACTS)

a) Base case without line compensation

Steady-state stability limit

$$P_{max} = \frac{V_1 V_2}{X} = 3.33 \text{ pu}$$

[0.5 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max}}\right) = 36.87°$$

[0.5 mark]

Critical clearing angle is obtained by equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P \, d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max} \sin\delta - P) \, d\delta$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = \mathbf{71.77°}$$

[1 mark]

b) With a STATCOM

$I_{max} = 4.2$ p.u.

When the rated current capacity of the STATCOM is reached

$$\frac{4V^2}{X}\left(1 - \cos\frac{\delta_{lim3}}{2}\right) = VI_{max} \rightarrow \delta_{lim3} = 2 \times \cos^{-1}\left(1 - \frac{XI_{max}}{4V}\right) = 93.53°$$

[1 mark]

Beyond this point

$$P = \frac{V^2}{X}\sin\delta + \frac{VI_{max}}{2}\sin\frac{\delta}{2} = P_{max0}\sin\delta + \frac{I_{max}}{2}\sin\frac{\delta}{2}$$

as $V = 1.0$ p.u.

[1 mark]

Power angle at which maximum power is transferred can be determined as follows:

Model Answers

$$\frac{dP}{d\delta} = 0 \rightarrow P_{max0}\cos\delta + \frac{I_{max}}{4}\cos\frac{\delta}{2} = 0 \rightarrow 2P_{max0}\left(\cos\frac{\delta}{2}\right)^2 + \frac{I_{max}}{4}\cos\frac{\delta}{2} - P_{max0} = 0$$

[2 marks]

$$\delta_{Pmax} = 2 \times \cos^{-1}\left[\frac{-\frac{I_{max}}{4} \pm \sqrt{\left(\frac{I_{max}}{4}\right)^2 + 8{P_{max0}}^2}}{4P_{max0}}\right] = 101.5°$$

Ignoring the other solution 284.41°

[1 mark]

As δ_{Pmax} is larger than δ_{lim3} the maximum power is given by

$$P_{max3} = P_{max0}\sin\delta_{Pmax} + \frac{I_{max}}{2}\sin\frac{\delta_{Pmax}}{2} = \mathbf{4.89\ p.u}$$

[2 marks]

c) With an SVC

Steady-state stability limit with susceptance limit

$$P_{max2} = \frac{V_1 V_2}{X/2} = 6.67 \text{ pu}$$

Initial power angle $\delta_0$is given by:

$$\delta_0 = 2 \times \sin^{-1}\left(\frac{P}{P_{max2}}\right) = 34.92°$$

[1 mark]

$B_{max} = 4.75$ p.u.

When the reactive capacity of the SVC is reached

$$\frac{4}{X}\left(1 - \cos\frac{\delta_{lim}}{2}\right) = V^2 B_{max} \rightarrow \delta_{lim} = 2 \times \cos^{-1}\left(1 - \frac{XB_{max}}{4}\right) = 99.86°$$

[1 mark]

Note $\delta_{lim} > 90°$ which dictates the following steps

[1 mark]

Model Answers

After the reactive capacity of the SVC is reached i.e. for $\delta > \delta_{lim}$ the power angle characteristics is given by

$$P = \frac{4}{X(4 - XB_{max})} \sin\delta = P_{max-lim} \sin\delta$$

$$P_{max-lim} = \frac{4}{0.4(4 - 0.4 \times 3.2)} = 5.19 \text{ p.u.}$$

[1 mark]

$$\delta_{max} = \pi - \sin^{-1}\left(\frac{P}{P_{max-lim}}\right) = 157.28°$$

[1 mark]

Equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P \, d\delta = \int_{\delta_{cr}}^{\delta_{lim}} \left(P_{max2} \sin\frac{\delta}{2} - P\right) d\delta + \int_{\delta_{lim}}^{\delta_{max}} (P_{max-lim} \sin\delta - P) \, d\delta$$

[2 marks]

$$\delta_{cr} = 2 \times \cos^{-1}\left[\frac{P(\delta_{max} - \delta_0) + 2P_{max2}\cos\frac{\delta_{lim}}{2} - P_{max-lim}(\cos\delta_{lim} - \cos\delta_{max})}{2P_{max2}}\right] = \mathbf{95.5°}$$

[1 marks]

d) With series compensation

With the TCSC the new steady-state stability limit would be

$$P_{max1} = \frac{V_1 V_2}{X(1-k)} = 4.76 \text{ pu}$$

[1 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 24.83°$$

[1 mark]

Critical clearing angle is

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = \mathbf{87.26°}$$

[1 mark]

Model Answers

Question 2 on System Operation with Low Inertia

a)

i)

Rate of Change of Frequency (RoCoF), the instant immediately after the system disturbance
System inertia, disturbance
[1 mark]

Frequency Nadir, the instant when the frequency deviation reaches a maximum value
Inertia, PRF, disturbance (EFR, damping)
[1 mark]

Steady-state Frequency, the period after the frequency reaches the steady-state
PFR, disturbance (EFR, damping)
[1 mark]

ii)

1)
$P_L = 1.53$,
[2 marks]

2)
$g(R, EFR) = (3 - 0.2EFR)R - 3(P_L - EFR)^2$,
$\frac{\partial g}{\partial R} = 3 - 0.2EFR = 2.9$,
[1 mark]

$\frac{\partial g}{\partial EFR} = -0.2R + 6(P_L - EFR) = 5.96$,
[1 mark]

$-\frac{\partial g}{\partial P_L} = 6(P_L - EFR) = 6.18$,
[1 mark]
Per unit of P_L reduction is equivalent to increasing 6.18/5.96=1.04 unit of EFR or increasing 6.18/2.9=2.13 unit of PFR.
[1 mark]
This is because P_L has the faster response time (instantaneously) than EFR (0.64s) and EFR than PFR (9.6s).
[1 mark]

b)

i) BOOKWORK. Wind turbines control their power intake and their speed using two actuators: the torque of the generator, which is controlled using power electronic converter, and the blade pitch angle, which is controlled using a servomotor.
Headroom can be created by using either actuator. A greater blade pitch angle could be kept in place at all times to be able to increase the torque by reducing the angle at any time.

Model Answers

At low and medium speeds, the torque could be used to stabilise the speed at an operating point with a higher tip-speed ratio than the one that corresponds to the maximum power point. This would enable the turbine to boost the output power by increasing the torque, which would both extract kinetic energy from the turbine to deliver power to the grid and increase the power intake from the wind on the longer term by moving closer to the maximum power point.

[4 marks]

ii) The equation of the synthetic inertia is based on the swing equation:

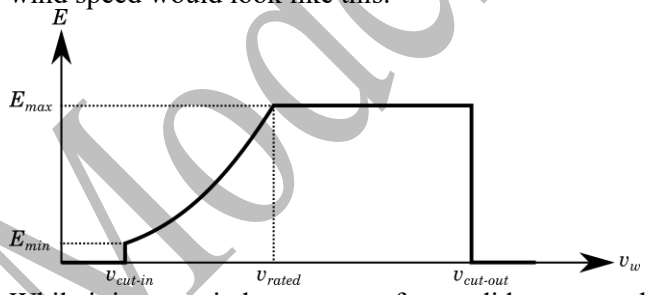
$$\Delta P = \frac{-2H}{f_{nominal}} S_{base} \frac{df}{dt}$$

This can be used to calculate how much the power output of the generator would change for a given RoCoF. A positive RoCoF would cause the power to reduce, which in terms of its implementation in a wind turbine it would imply a reduction of the power output, which can be achieved by using speed and blade pitch angle control. A negative RoCoF would cause the power to increase, which requires the generator and its power converter to be sized for this increment. In this problem we know that our converter can handle an increase of power output of 1 MW, we can use this figure to calculate the highest H it would be able to emulate:

$$H = -\frac{\Delta P \, f_{nominal}}{2 \, S_{base} \frac{df}{dt}} = -\frac{1 \text{ MW} \cdot 50 \text{ Hz}}{2 \cdot 8 \text{ MW} \cdot (\text{-}2 \text{ Hz/s})} = 1.56 \, s$$

[3 marks]

iii) The turbine is at rest for wind speeds below cut-in or above cut-out. The turbine will spin at a speed that is linearly proportional to the wind speed under the Maximum Power Point Tracking (MPPT) region when the output power yield is below rated power. The turbine will operate at constant speed at any wind speed between rated speed and cut-out. The kinetic energy of the turbine is proportional to the square of the rotating speed, therefore the kinetic energy versus wind speed would look like this:



While it is not strictly necessary for candidates to explicitly say that the energy will increase with the square of the wind speed in the MPPT region, they are expected to produce a graph similar to the one shown below and to justify the approximate shape of the curve considering the behaviour of the turbine under the different ranges of wind speed.

[3 marks]

Model Answers

Question 3 on Power System State Estimation

a)

i) What is the purpose of state estimation in power systems? [4]

The network control centre needs to be aware of operational status at any time such as how much is the flow in any branch, node voltage magnitudes, how many circuits are under operation and what is the extent of their loading, whether volts amps and flows are within the limit of the designed capacity of the equipment etc. If some feeders are lost whether the rest of the network will be secured -that can only be done through security analysis which needs estimated state. The states cannot be obtained with measurement only as not every node and branch are measured, neither all the quantities can be easily measured. There are errors associated with measurements also. It is required to obtain accurate value of the quantities to synthesise other quantities those are not measured yet important for the operator to know. All these circumstances and necessity lead to the requirement of a state estimation.

[4 marks]

ii) Why can't power flow replace the state estimation? [3]

Power flow requires precise value of the demands, generation, and node voltages – all of which are measured. So, power flow cannot be substituted.

[3 marks]

iii) How do the errors differ from bad data in a state estimation? [3]

Here the student is expected to elaborate or define what are errors associated with measurement which relates metering class, accuracy, sensitivity etc. On the bad data it is expected that they will point on gross telemetering errors, impact of data manipulation, RTU errors etc.

[3 marks]

b) In the context of power distribution system

i) What is real measurement? [2]

I am expecting students to write about real amps, volt and power measurements through analogue and digital devices

[2 marks]

ii) What is pseudo measurement? [2]

This is where real measurements are not available but forecast is used or previous historical information is taken advantage of to generate one . I am expecting the student to list few of them such as unmonitored substation , feeder loads etc

Model Answers

[2 marks]

iii) What is zero injection? [2]

This is where there is a node but no real generation or demand connected so what comes in that goes out ignoring losses , a transformer bus is an ideal example

[2 marks]

(iv) How are these measurements represented in state estimation objective function? [4]

The students are expected to characterise each type of measurements through statistical attributes and range. Then connect those measurements to objective function in order of their importance and trustworthiness. Relative weights associated to each measurement is very much needed in the answer, shifting some very accurate measurement as constraints of the optimisation problem is another. Etc ..

[4 marks]

Model Answers

Question 4 on Electricity Market and Peer-to-Peer Trading

a)

i) Since the wind farm "O1" is operating at 70% of its capacity, the power dispatch of the wind farm is $400 \times 70\% = 280$ MW. Then, the supply-demand curve can be constructed as follow:

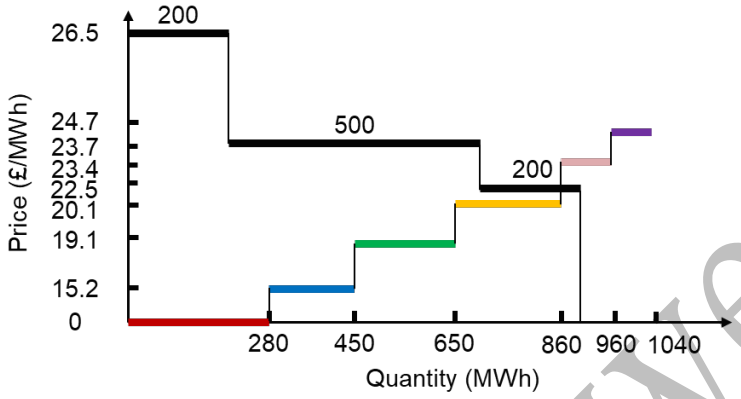


Figure 1: Market demand-supply curve for perfect competition case.

[1 mark]

Generation dispatches of 6 units and demand schedules of 3 demands:
Generation of S1 = 170 MW
Generation of S2 = 0 MW
Generation of S3 = 200 MW
Generation of O1 = 280 MW
Generation of O2 = 210 MW
Generation of O3 = 0 MW
Demand of D1 = 200 MW
Demand of D2 = 500 MW
Demand of D3 = 160 MW

[1 mark]

Market-clearing price = cost of unit O2 = 20.1 £/MWh.

[1 mark]

Generation profits of 6 units and demand benefits of 3 demands:
Profit of S1 = 170 MW × (20.1 – 15.2) £/MWh × 1 hour = £833
Profit of S2 = £0
Profit of S3 = 200 MW × (20.1 – 19.1) £/MWh × 1 hour = £200
Profit of O1 = 280 MW × (20.1 – 0) £/MWh × 1 hour = £5,628
Profit of O2 = 210 MW × (20.1 – 20.1) £/MWh × 1 hour = £0
Profit of O3 = £0
Benefit of D1 = 200 MW × (26.5 – 20.1) £/MWh × 1 hour = £1,280

Model Answers

Benefit of D2 = 500 MW × (23.7 – 20.1) £/MWh × 1 hour = £1,780
Benefit of D3 = 160 MW × (22.5 – 20.1) £/MWh × 1 hour = £384
[1 mark]

Social welfare = demand benefits + generation profits = £(833 + 0 + 200 + 5628 + 0 + 0) + £(1,280 + 1780 + 384) = £6,661 + £3,444 = £10,105
[1 mark]

ii) The strategic producer owing units S1-S3 offers strategically at 1.2 times their true cost, so the operational details of generation units can be updated as:

Table 1: Operational details of generation units

Unit	S1	S2	S3	O1	O2	O3
Technology	Coal	Gas	Gas	Wind	Gas	Gas
Bidding Cost (£/MWh)	**18.24**	**28.08**	**22.92**	0	20.1	24.7
Cost (£/MWh)	15.2	23.4	19.1	0	20.1	24.7
Capacity (MW)	170	100	200	400	210	80

Then, the new supply-demand curve can be constructed as follow:

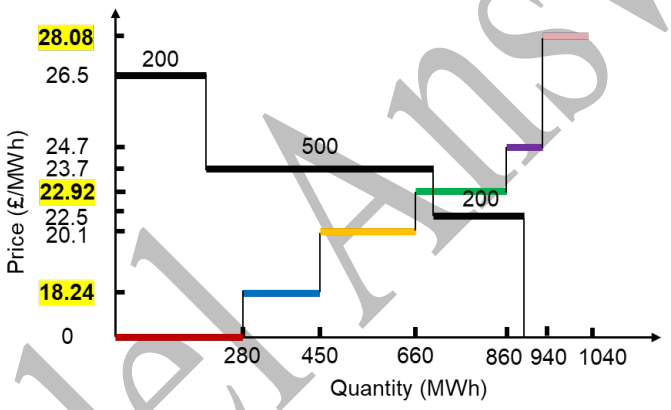


Figure 2: Market demand-supply curve for strategic offering case.
[1 mark]

Generation dispatches of 6 units and demand schedules of 3 demands:
Generation of S1 = 170 MW
Generation of S2 = 0 MW
Generation of S3 = 40 MW
Generation of O1 = 280 MW
Generation of O2 = 210 MW
Generation of O3 = 0 MW
Demand of D1 = 200 MW
Demand of D2 = 500 MW
Demand of D3 = 0 MW
[1 mark]

Market-clearing price = strategic bidding cost of unit S3 = 22.92 £/MWh.

Model Answers

[1 mark]

Generation profits of 6 units and demand benefits of 3 demands:
Profit of S1 = 170 MW × (22.92 – 15.2) £/MWh × 1 hour = £1,312.4
Profit of S2 = £0
Profit of S3 = 40 MW × (22.92 – 19.1) £/MWh × 1 hour = £152.8
Profit of O1 = 280 MW × (22.92 – 0) £/MWh × 1 hour = £6,417.6
Profit of O2 = 210 MW × (22.92 – 20.1) £/MWh × 1 hour = £592.2
Profit of O3 = £0
Benefit of D1 = 200 MW × (26.5 – 22.92) £/MWh × 1 hour = £2,716
Benefit of D2 = 500 MW × (23.7 – 22.92) £/MWh × 1 hour = £390
Benefit of D3 = £0
[1 mark]

Social welfare = demand benefits + generation profits = £(1,312.4 + 0 + 152.8 + 6,417.6 + 592.2 + 0) + £(2,716 + 390 + 0) = £8,475 + £3,106 = £11,581
[1 mark]

b)

i. Based on Table 3, the supply-demand curve of 3 consumers and 3 prosumers for the double-auction market can be constructed as follow:

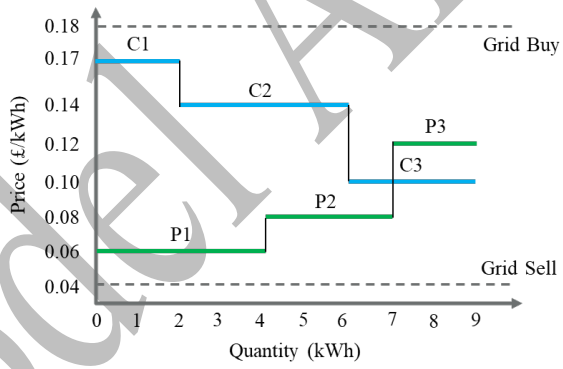


Figure 3: Demand-supply curve for double-auction market.

[1 mark]

First Trading Activity happens between C1 and P1:
Trading price = (0.17 + 0.06)/2 = 0.115 £/kWh
Trading quantity = min(2, 4) = 2 kWh
Updated order book

Consumer/Prosumer	C2	C3	P1	P2	P3
Bid/Offer price (£/kWh)	0.14	0.10	0.06	0.08	0.12
Quantity (kW)	4	3	2	3	2

[1 mark]

Model Answers

Second Trading Activity happens between C2 and P1:
Trading price = (0.14 + 0.06)/2 = 0.1 £/kWh
Trading quantity = min(4, 2) = 2 kWh
Updated order book

Consumer/Prosumer	C2	C3	P2	P3
Bid/Offer price (£/kWh)	0.14	0.10	0.08	0.12
Quantity (kW)	2	3	3	2

[1 mark]

Third Trading Activity happens between C2 and P2:
Trading price = (0.14 + 0.08)/2 = 0.11 £/kWh
Trading quantity = min(2, 3) = 2 kWh
Updated order book

Consumer/Prosumer	C3	P2	P3
Bid/Offer price (£/kWh)	0.10	0.08	0.12
Quantity (kW)	3	1	2

[1 mark]

Fourth Trading Activity happens between C3 and P2:
Trading price = (0.10 + 0.08)/2 = 0.09 £/kWh
Trading quantity = min(3, 1) = 1 kWh
Updated order book

Consumer/Prosumer	C3	P3
Bid/Offer price (£/kWh)	0.10	0.12
Quantity (kW)	2	2

[1 mark]

ii. If there is no local energy market, 3 consumers purchase electricity from the main grid at grid buy price 0.18 £/kWh and 3 prosumers sell electricity to the main grid at grid sell price 0.04 £/kWh. The cost of 3 consumers and revenue of 3 prosumers can be calculated as:

Cost of C1 = 2 kW × 0.18 £/kWh × 1 hour = £0.36
Cost of C2 = 4 kW × 0.18 £/kWh × 1 hour = £0.72
Cost of C3 = 3 kW × 0.18 £/kWh × 1 hour = £0.54
Revenue of P1 = 4 kW × 0.04 £/kWh × 1 hour = £0.16
Revenue of P2 = 3 kW × 0.04 £/kWh × 1 hour = £0.12
Revenue of P3 = 2 kW × 0.04 £/kWh × 1 hour = £0.08
Total Cost of 3 consumers = £0.36 + £0.72 + £0.54 = £1.62
Total Revenue of 3 prosumers = £0.16 + £0.12 + £0.08 = £0.36
[2 marks]

If there is a local energy market, four local trading activity occur, as discussed in (c). The cost of 3 consumers and revenue of 3 prosumers can be calculated as:
Cost of C1 = 2 kWh × 0.115 £/kWh × 1 hour = £0.23
Cost of C2 = (2 kWh × 0.1 £/kWh + 2 kWh × 0.11 £/kWh) × 1 hour = £0.42

Model Answers

Cost of C3 = (1 kWh × 0.09 £/kWh + 1 kWh × 0.18 £/kWh) × 1 hour = £0.27
Revenue of P1 = (2 kWh × 0.115 £/kWh + 2 kWh × 0.1 £/kWh) × 1 hour = £0.43
Revenue of P2 = (2 kWh × 0.11 £/kWh + 1 kWh × 0.09 £/kWh) × 1 hour = £0.31
Revenue of P3 = 2 kW × 0.04 £/kWh × 1 hour = £0.08
Total Cost of 3 consumers = £0.23 + £0.42 + £0.27 = £0.92
Total Revenue of 3 prosumers = £0.43 + £0.31 + £0.08 = £0.82
[2 marks]

In this local energy market, 7 kWh of local trading occurs, indicating a reduction in dependence on the upstream grid. Additionally, the local energy market results in a decrease in costs for 3 consumers by £0.7 (£1.62 - £0.92) and an increase in revenue for 3 prosumers by £0.46 (£0.82 - £0.36).
[1 mark]

Model Answers